CS24305 — THEORY (3.0 CR) | MODULE II NOTES | BIT MESRA | NEP SCHEME

Module 2: Physical Media & Signal Encoding

Guided/Wireless Media, NRZ, Manchester, QAM & PCM

Module II: Transmission Media & Signal Encoding — Topics Covered

1. Guided Media (Twisted Pair, Coax, Optical Fiber)
2. Digital Signal Encoding: NRZ, Manchester, AMI
3. Analog Modulation: ASK, FSK, PSK, QAM
4. Pulse Code Modulation (PCM) & Quantization

1. Digital Signal Encoding Techniques

Encoding Scheme	Rule / Signal Transition	Key Advantages & Drawbacks
NRZ-L (Level)	Bit 0 = High level, Bit 1 = Low level	Simple; susceptible to DC baseline wander and clock drift on long runs of 0s/1s.
NRZI (Invert)	Bit 1 = Transition at start; Bit 0 = No transition	Solves sync for consecutive 1s, but fails on strings of 0s.
Bipolar-AMI	Bit 0 = Zero voltage; Bit 1 = Alternating +V and -V	Zero DC component; easy error detection; fails sync on consecutive 0s.
Manchester (Ethernet)	Bit 0 = High-to-Low transition; Bit 1 = Low-to-High transition	**Self-clocking** (transition at middle of every bit); requires $2\times$ bandwidth.
Diff. Manchester	Transition in middle; Bit 0 = Transition at start; Bit 1 = No transition at start	Differential noise immunity; self-clocking; standard in Token Ring (IEEE 802.5).

2. Analog Modulation & 16-QAM Constellation

Quadrature Amplitude Modulation (QAM) modulates both amplitude and phase simultaneously. In 16-QAM, each transmitted constellation point represents **4 bits** $(\log_2(16) = 4)$, enabling high spectral efficiency.